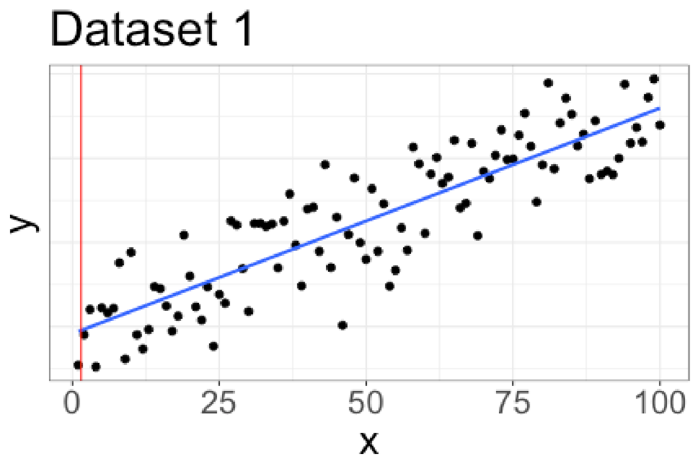
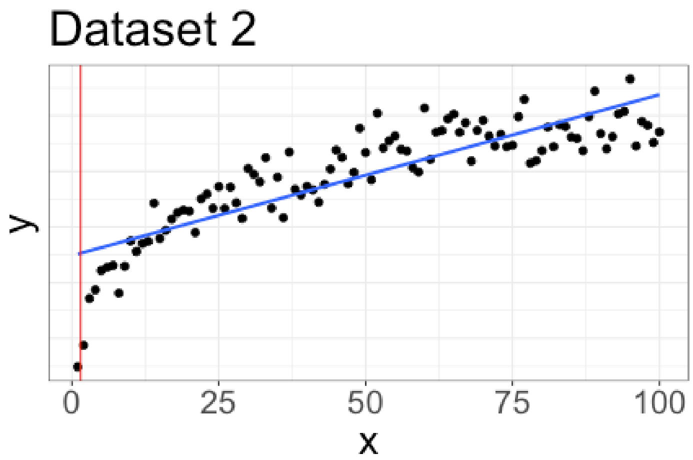
Assumptions and diagnostics

Predictions revisited

Here are two different datasets with similar correlation ($r \approx 0.8$):

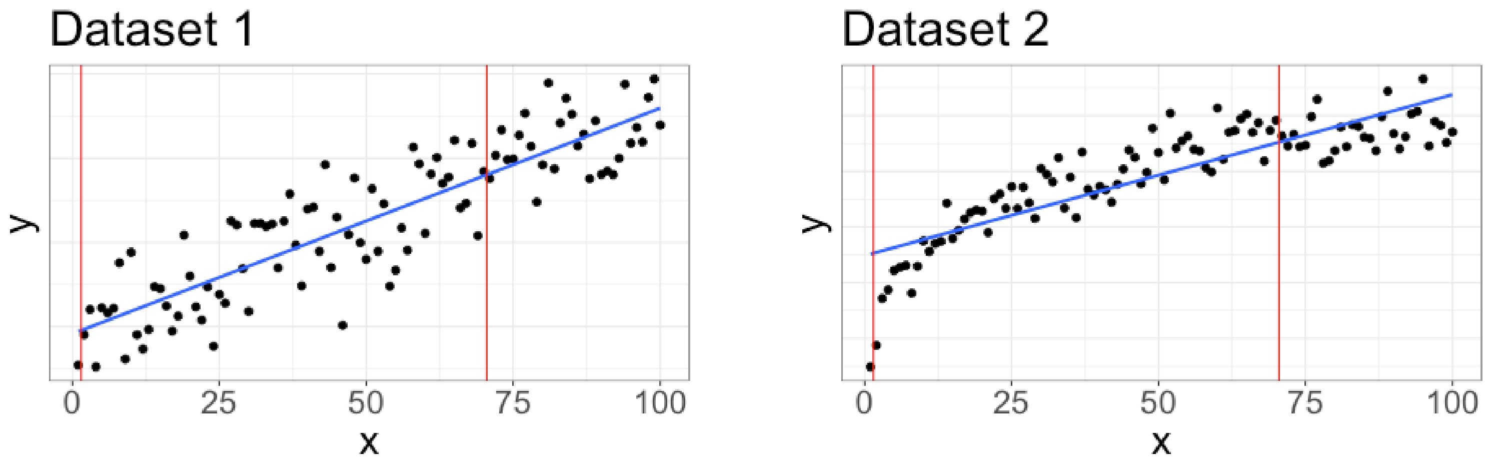




For each dataset, suppose I wish to make a prediction for $x = 2$. For which dataset would I expect to predict better?

Predictions revisited

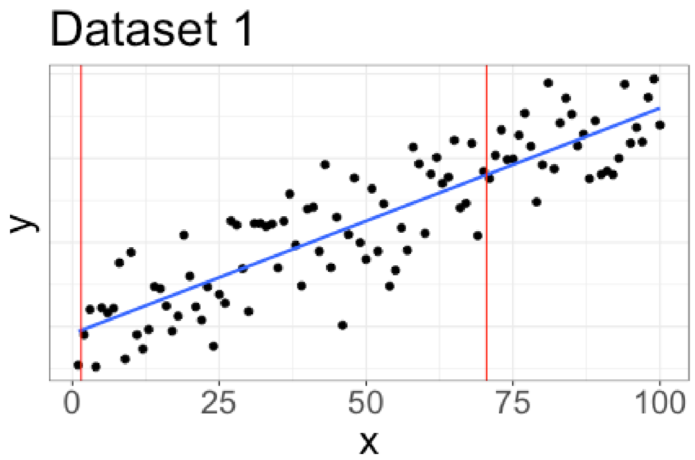
Here are two different datasets with similar correlation ($r \approx 0.8$):

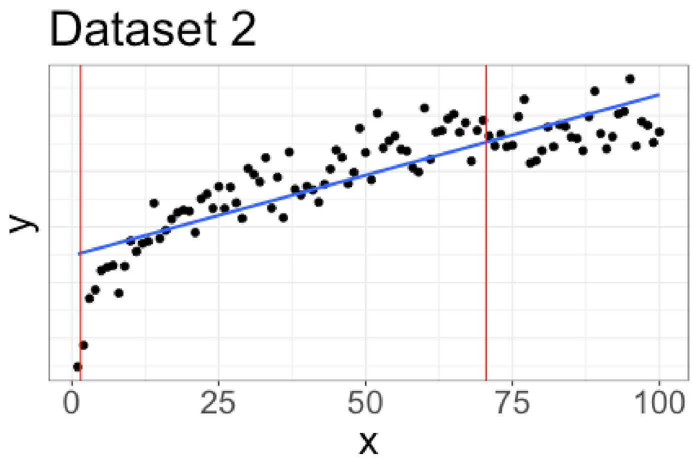


> For each dataset, suppose I wish to make a prediction for $x = 70$. For which dataset would I expect to predict better?

Predictions revisited

Here are two different datasets with similar correlation ($r \approx 0.8$):

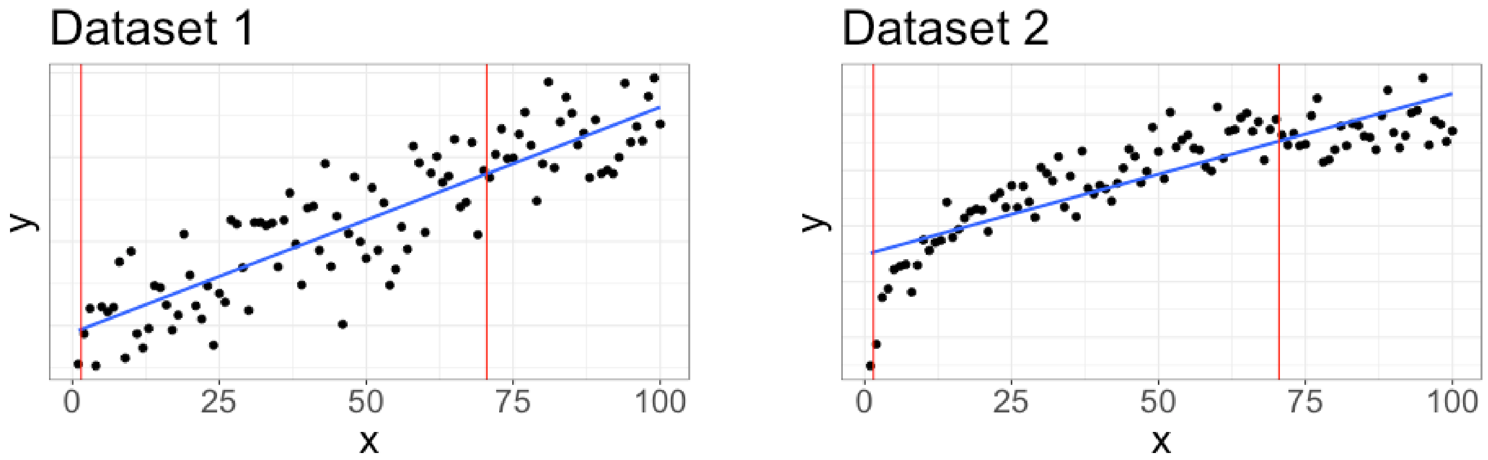




> For which dataset does the line appear to be a better fit?

Predictions and linearity

If we use a line to describe non-linear data, the quality of our predictions may depend on the value of x:



Predictions revisited

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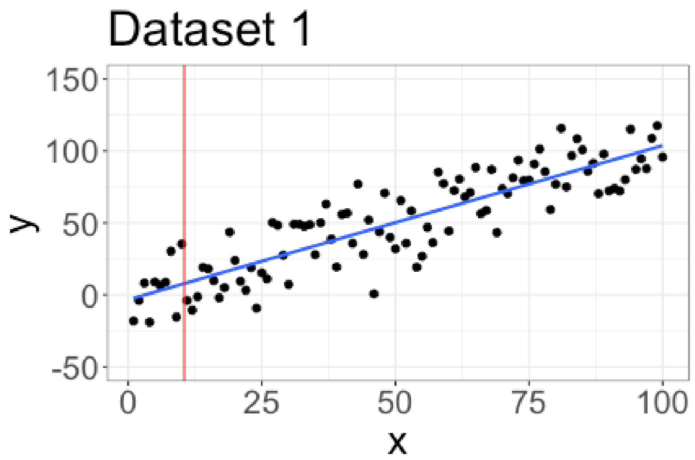
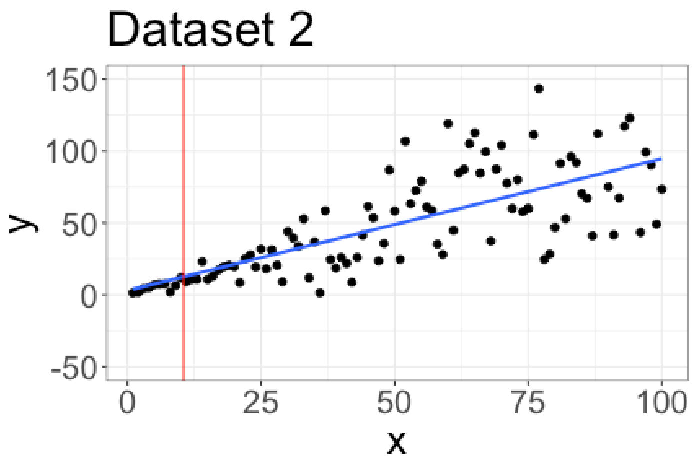
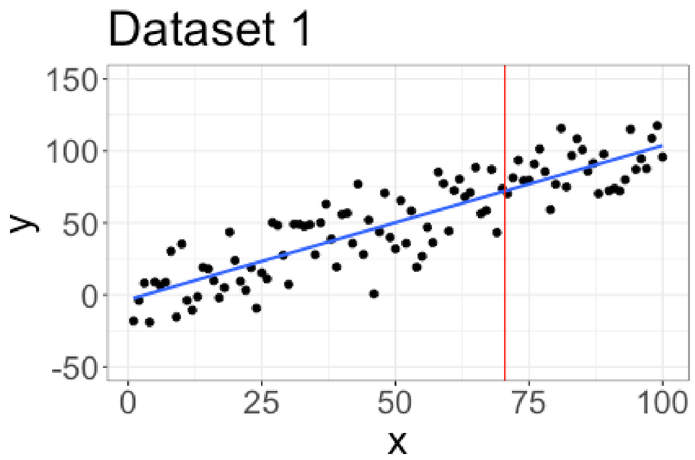


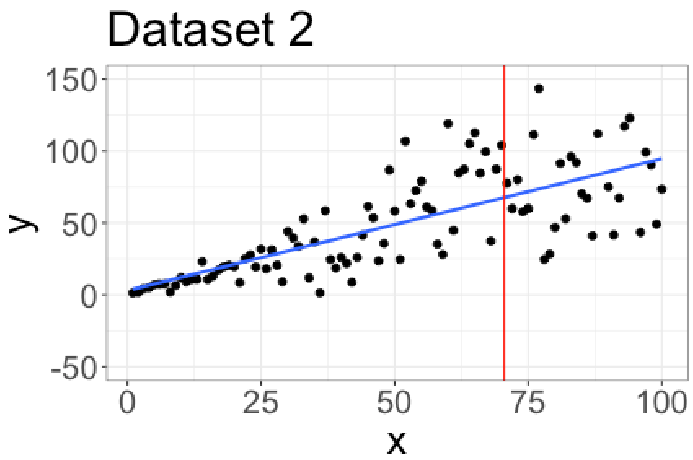


> For each dataset, suppose I wish to make a prediction for $x = 10$. For which dataset would I expect to predict better?

Predictions revisited

Here are two different datasets with similar correlation ($r \approx 0.8$):

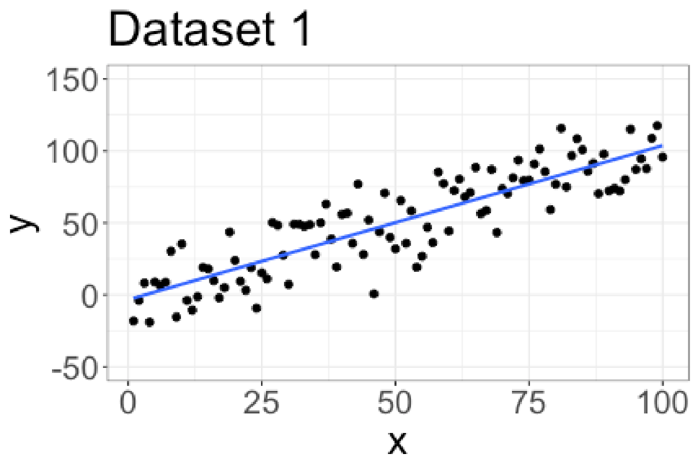


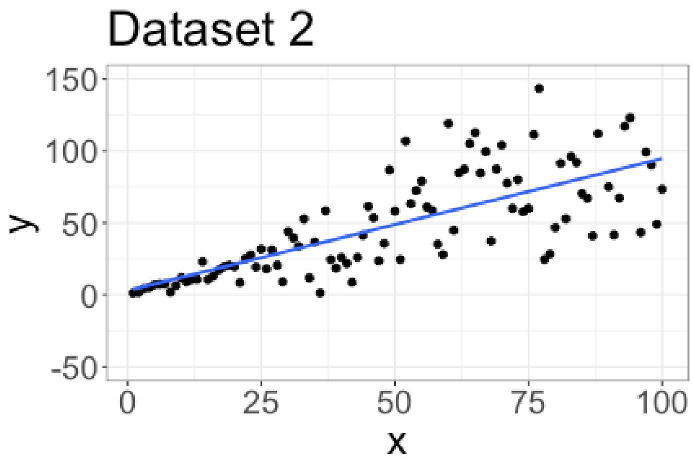


> For each dataset, suppose I wish to make a prediction for $x = 70$. For which dataset would I expect to predict better?

Predictions revisited

Here are two different datasets with similar correlation ($r \approx 0.8$):

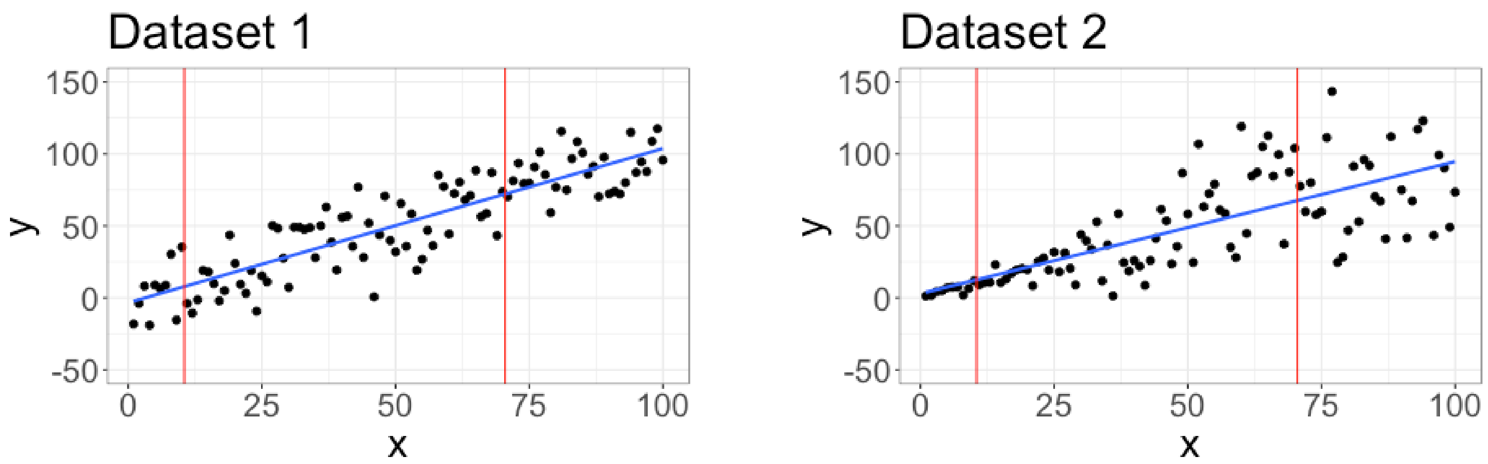




> Does a linear fit seem appropriate for both datasets?

Predictions and variability

If the variability of our residuals changes for different values of x, the quality of our predictions depends on x:



Common linear model assumptions

$$Y = \beta_0 + \beta_1 X + \varepsilon$$

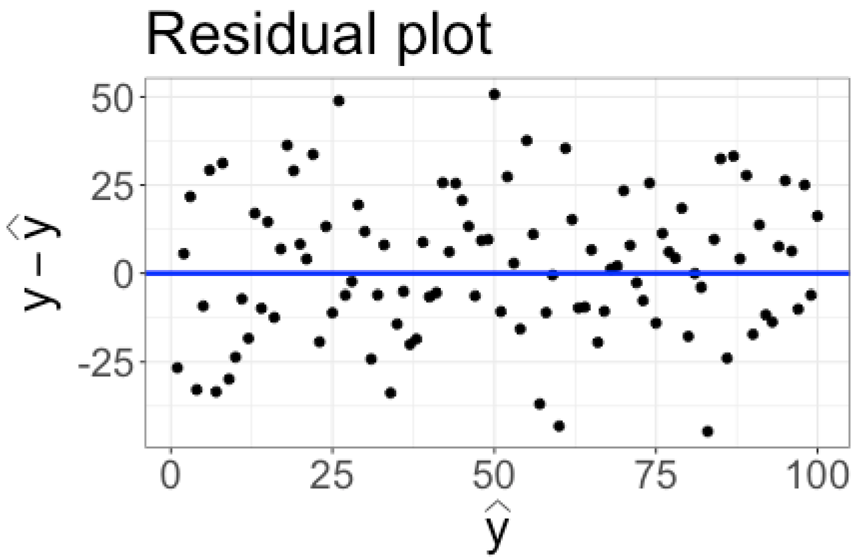
Shape assumption: the shape of the regression model is at least approximately correct

- When we fit linear regression, we assume the relationship is approximately linear

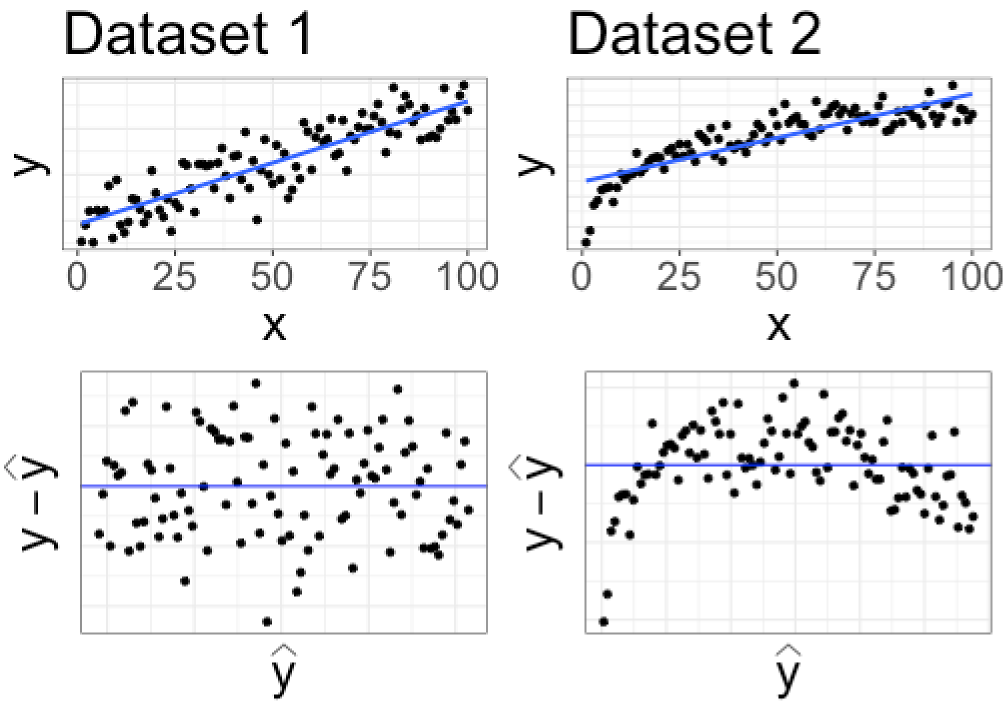
Constant variance assumption: Variance of the noise ε is the same for all values of the predictor X

Assessing shape and constant variance: residual plots

Residual plot: plot residuals $y - \hat{y}$ on vertical axis, and predicted values $\hat{y}$ on horizontal axis. Add a horizontal line at 0.



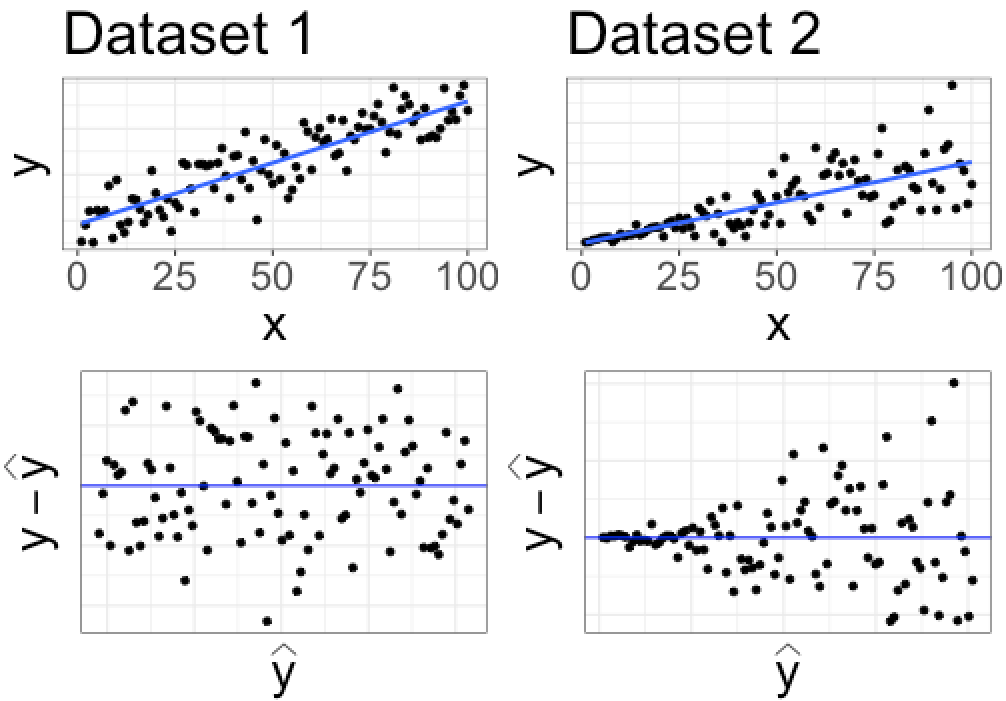
Residual plots



Shape assumption is reasonable if:

- residuals appear to be scattered randomly above and below 0
- no clear patterns in the residuals

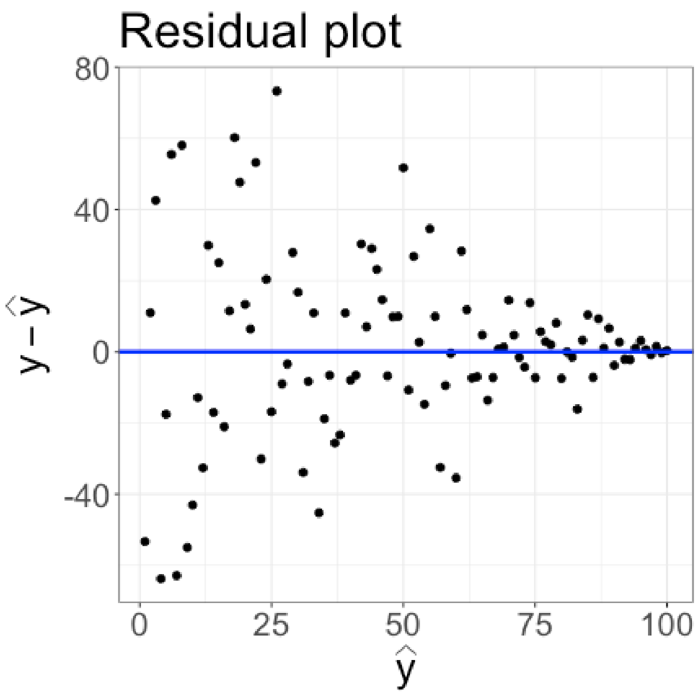
Residual plots



Constant variance assumption is reasonable if:

- the band of residuals has relatively constant width

Concept check



Which assumption looks reasonable?

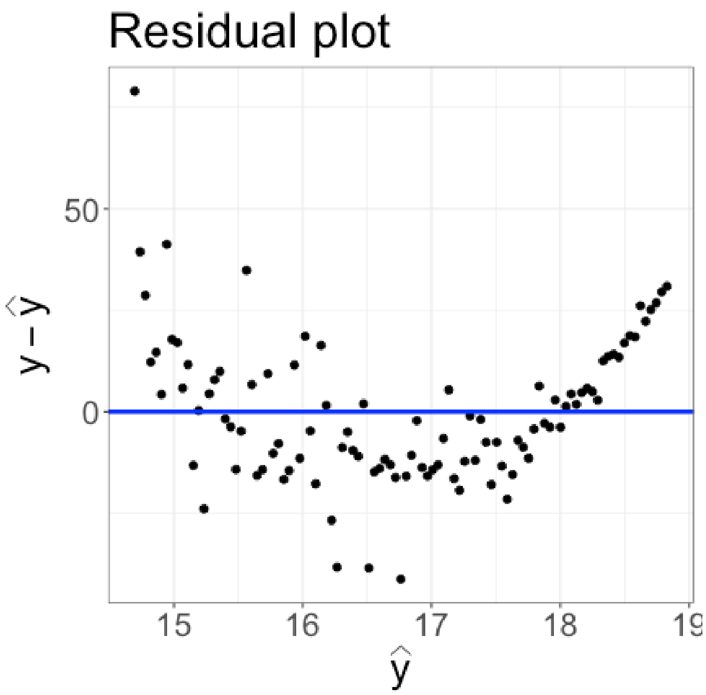
(A) Shape

(B) Constant variance

(C) Both shape and constant variance

(D) Neither shape nor constant variance

Concept check



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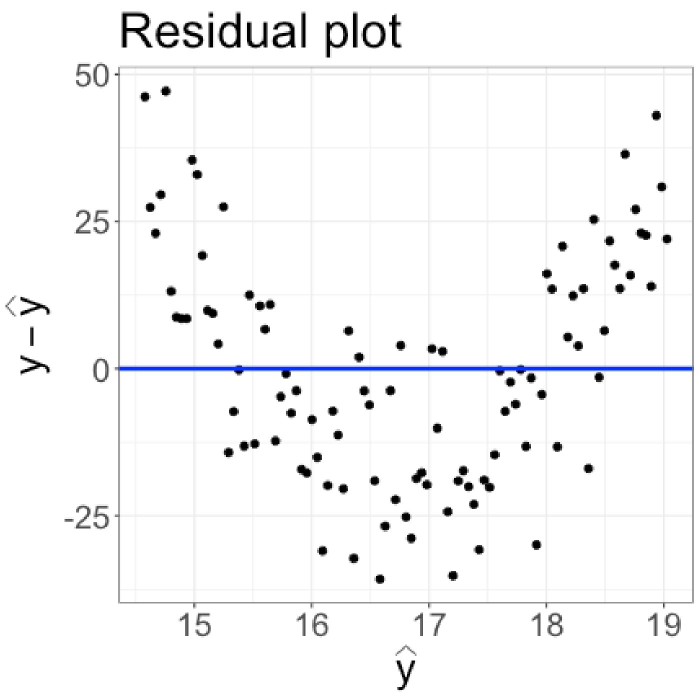
Concept check



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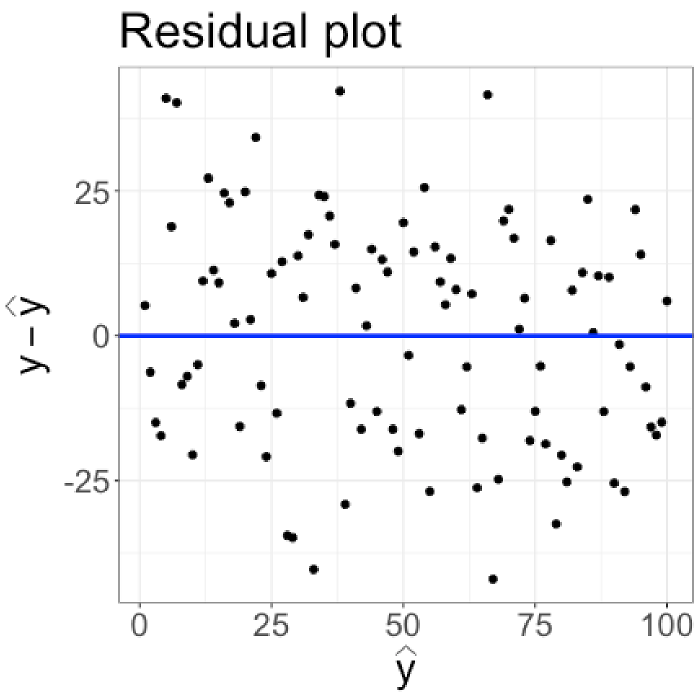
Concept check



Which assumption looks reasonable?

(A) Shape

(B) Constant variance

(C) Both shape and constant variance

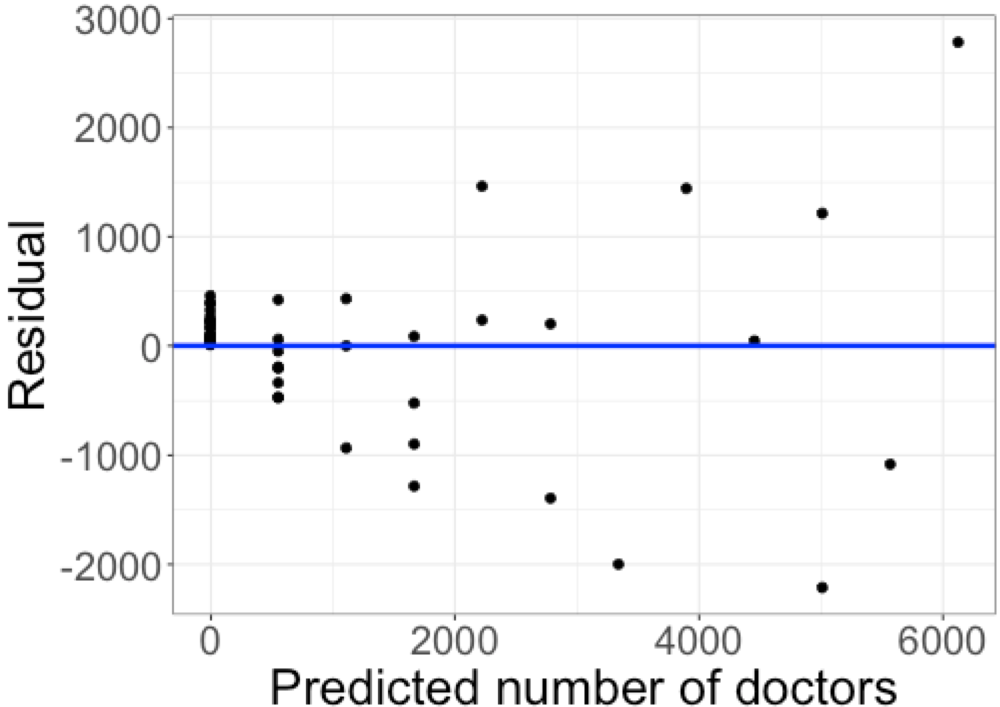
(D) Neither shape nor constant variance

Class activity

https://sta112-s26.github.io/class_activities/ca_08.html

- Evaluate shape and constant variance assumptions with residual plots
- Submit HTML at end of class

Class activity



> Do the shape and constant variance assumptions look reasonable?